

Question	Answer	Marks
6(a)(i)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	B1
6(a)(ii)	charge	B1
6(b)(i)1.	$E = I^2 R t$ or $E = V I t$ or $E = (V^2 / R) t$	C1
	$E = 2.5^2 \times 2.0 \times 5.0 \times 60$ or $5.0 \times 2.5 \times 5.0 \times 60$ or $(5.0^2 / 2.0) \times 5.0 \times 60$ $= 3800 \text{ J}$	A1
6(b)(i)2.	p.d. = $8.0 - (2.0 \times 2.5)$ $= 3.0 \text{ V}$	A1
6(b)(ii)	$I_X = 3.0 / 15 = 0.20 \text{ (A)}$	C1
	$I_Y = 2.5 - 0.20 = 2.3 \text{ (A)}$	C1
	$R_Y = 3.0 / 2.3$ $= 1.3 \Omega$	A1
	or	
	$R_T = 3.0 / 2.5 = 1.2 (\Omega)$ or $(8.0 / 2.5) - 2.0 = 1.2 (\Omega)$	(C1)
	$1 / 1.2 = 1 / 15 + 1 / R_Y$	(C1)
	$R_Y = 1.3 \Omega$	(A1)

Question	Answer	Marks
6(b)(iii)1.	Z has larger radius/diameter/(cross-sectional) area	B1
	Z has (material of) smaller resistivity/greater conductivity	B1
6(b)(iii)2.	current/ I (in battery) increases	M1
	($P = EI$ so) power/ P (produced by battery) increases	A1

Question	Answer	Marks
7(a)		B1